

Parent Function Families

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Parent function: the _____ form of a family — no shifts, stretches, or reflections.

Domain / Range: domain is the set of allowed _____-values; range is the set of resulting _____-values.

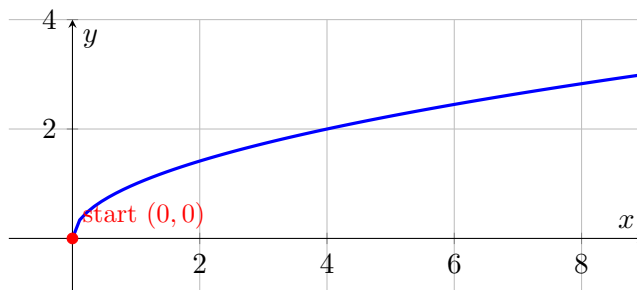
We do together. Name the parent of $f(x) = \sqrt{x}$ and give its domain.

Step 1. Match the rule to a known family: this is the _____ parent.

Step 2. Recall its shape: it starts at the point _____ and curves up to the right.

Step 3. A square root needs a non-negative inside, so the domain is _____.

See it:



$y = \sqrt{x}$ — starts at the origin, domain $x \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Name the parent of $f(x) = \frac{1}{x}$ and give its domain.

b) Name the parent of $f(x) = |x|$ and give its range.

Summary (fill in): To name a parent: match the _____ to a family, recall its _____, then state the _____.